Complete schematic - sheet 3

Every component terminal is shown. Equal net labels are connected; use connection IDs in the CSV for physical routing.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

ESP1 | ESP32-WROOM-32 DevKit

INVENTORY / VERIFY

module terminal symbol

5V_VIN	V5_MCU
3V3	V3V3
GND	GND
GPIO16	L_FWD_3V3
GPIO17	L_REV_3V3
GPIO18	R_FWD_3V3
GPIO19	R_REV_3V3
GPIO14	ENABLE_3V3
GPIO21	BRUSH_IN1
GPIO22	BRUSH_IN2
GPIO23	CONVEYOR_IN3
GPIO25	CONVEYOR_IN4
GPIO32	TF_3V3
GPIO34	EF_SAFE
GPIO33	TL_3V3
GPIO35	EL_SAFE
GPIO13	TR_3V3
GPIO36	ER_SAFE
GPIO27	ESTOP_OK
GPIO0	NC - INSULATE
GPIO2	NC - INSULATE
GPIO4	NC - INSULATE
GPIO5	NC - INSULATE
GPIO12	NC - INSULATE
GPIO15	NC - INSULATE
GPIO26	NC - INSULATE
GPIO39	NC - INSULATE
UART0_TX_GPIO1	NC - INSULATE
UART0_RX_GPIO3	NC - INSULATE
EN_RESET	NC - INSULATE

US_L | HC-SR04 left

INVENTORY / VERIFY

module terminal symbol

VCC	V5_LOGIC
TRIG	TL_5V
ECHO	EL_RAW
GND	GND

U1 | SN74AHCT244N 3.3-to-5 V buffer

ADDITIONAL / MISSING OR SELECT

SN74AHCT244N

1 / 1OE_N	GND
2 / 1A1	L_FWD_3V3
3 / 2Y4	TR_5V
4 / 1A2	L_REV_3V3
5 / 2Y3	TL_5V
6 / 1A3	R_FWD_3V3
7 / 2Y2	TF_5V
8 / 1A4	R_REV_3V3
9 / 2Y1	ENABLE_5V
10 / GND	GND
11 / 2A1	ENABLE_3V3
12 / 1Y4	R_REV_5V
13 / 2A2	TF_3V3
14 / 1Y3	R_FWD_5V
15 / 2A3	TL_3V3
16 / 1Y2	L_REV_5V
17 / 2A4	TR_3V3
18 / 1Y1	L_FWD_5V
19 / 2OE_N	GND
20 / VCC	V5_LOGIC

DRV_A | ONE L298N: A brushes / B conveyor

INVENTORY / VERIFY

module terminal symbol

Vs_12V	AUX_SUPPLY
5V_LOGIC	V5_LOGIC
GND	GND
IN1	BRUSH_IN1
IN2	BRUSH_IN2
IN3	CONVEYOR_IN3
IN4	CONVEYOR_IN4
ENA	EN_DRIVER_5V
ENB	EN_DRIVER_5V
OUT1	DRV_A_OUT1
OUT2	DRV_A_OUT2
OUT3	DRV_A_OUT3
OUT4	DRV_A_OUT4

DRV_L | BTS7960 / IBT-2 left

INVENTORY / VERIFY

module terminal symbol

B+	LEFT_SUPPLY
B-	GND
M+	DRV_L_MP
M-	DRV_L_MN
VCC	V5_LOGIC
GND	GND
RPWM	L_FWD_5V
LPWM	L_REV_5V
R_EN	EN_DRIVER_5V
L_EN	EN_DRIVER_5V
R_IS	NC - INSULATE
L_IS	NC - INSULATE

DRV_R | BTS7960 / IBT-2 right

INVENTORY / VERIFY

module terminal symbol

B+	RIGHT_SUPPLY
B-	GND
M+	DRV_R_MP
M-	DRV_R_MN
VCC	V5_LOGIC
GND	GND
RPWM	R_FWD_5V
LPWM	R_REV_5V
R_EN	EN_DRIVER_5V
L_EN	EN_DRIVER_5V
R_IS	NC - INSULATE
L_IS	NC - INSULATE

US_F | HC-SR04 front

INVENTORY / VERIFY

module terminal symbol

VCC	V5_LOGIC
TRIG	TF_5V
ECHO	EF_RAW
GND	GND

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.